

Name : Patel Dhruv Mukeshbhai

Roll No : H-14

Sub : Fullstack

Assingment 3

Question 1:

Please fill out the form below. All fields are required.

Errors found:

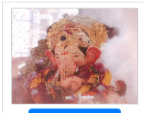
Confirm password is required.

Passwords do not match.

Username:	chandresh
Email Address:	2023040252@vnsgu.ac.in
Password:	*****
Confirm Password:	****
Gender:	☒ Male ☐ Female ☐ Other
Hobbies:	☒ Reading ☒ Music ☐ Coding ☐ Sports
Profile Picture (Single file):	Choose File No file chosen
Other Pictures (Multiple files):	Choose Files No file chosen
Register Reset	

Registration Details (Tabular View)

Success! The user has been registered successfully.

Username	chandresh
Email	2023040252@vnsgu.ac.in
Gender	Male
Hobbies	Reading, Music
Profile Picture	 Download
Other Pictures	 Download

← Register Another User

Question 2

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localhost:3002/dashboard

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Protected Route 1: Dashboard

[Dashboard \(Protected 1\)](#) | [Profile Details \(Protected 2\)](#) | [Logout](#)

Authentication Status: Active Session (File-Based Store)

Full Name	System Administrator
Username	admin
Role	Admin
Login Time	1/10/2026, 9:08:36 pm
Session ID	b6UVU8qw_1nFj0sDy139JveOmBwdgCDk
Page Refresh/Visit Count	3 times

* Note: Refreshing this page increments the session visit count persisted in the session file on the server.

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localhost:3002/profile

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Protected Route 2: User Profile

[Dashboard \(Protected 1\)](#) | [Profile Details \(Protected 2\)](#) | [Logout](#)

User Account	admin
Full Name	System Administrator
Email	admin@example.com
Role	Admin
Active Session ID	b6UVU8qw_1nFj0sDy139JveOmBwdgCDk

File Session Store Information:

The session file for this login is physically saved on the disk in:
C:\Users\admin\Desktop\assignment3_fullstack\Question_2\sessions\b6UVU8qw_1nFj0sDy139JveOmBwdgCDk.json

Even if the server restarts or memory is cleared, this session persists because it is stored in a JSON file on the disk!

Login Application (File Session Store)

Logged out successfully.

Username:	
Password:	
Login	

Demo Credentials:

1. Username: admin | Password: password123

2. Username: student | Password: student123

Question 3:

←↻📄localhost:3003/dashboard

Protected Route 1: Redis Dashboard

Dashboard (Protected 1)

 |

Redis Session Inspection (Protected 2)

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Logout

Authentication Status: Active Redis-backed Session

Username	redis_admin
Full Name	Redis Admin User
Role	Administrator
Login Time	1/10/2026, 11:01:05 pm
Session ID (Cookie)	EuJijlarxl_0XinswHb1H9ljeKTzgPVg
Redis Storage Key	student_sess:EuJijlarxl_0XinswHb1H9ljeKTzgPVg
Redis Engine Status	Embedded In-Memory Redis Mock (Zero setup needed)
Page Views (In Redis)	2 times

←↻📄localhost:3003/reports

Protected Route 2: Redis Session Store Inspection

Dashboard (Protected 1)

 |

Redis Session Inspection (Protected 2)

 |

Logout

User Account	redis_admin (Redis Admin User)
Session Cookie ID	EuJijlarxl_0XinswHb1H9ljeKTzgPVg
Key Queried in Redis	student_sess:EuJijlarxl_0XinswHb1H9ljeKTzgPVg
Redis Engine	Embedded In-Memory Redis Mock (Zero setup needed)

Question 4

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localhost:3004/employees?msg=Employee%20chandresh%20Patel%20(EMP1001)%20inserted%20successfully%20Password%20encrypted%20with%20bcrypt&emailLink=https%3A...

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ERP System - Admin Panel (Employees Management)

+ Add New Employee

Logout

Employee chandresh Patel (EMP1001) inserted successfully! Password encrypted with bcrypt.
Nodemailer Sent Email: [Click here to view the simulated sent email on Ethereal](#)

Employee Records (Mongoose & Net Salary Calculation)

Formula: Net Salary = Basic Salary + Allowances - Deductions

Emp ID	Name	Email	Department	Designation	Basic Salary	Allowances	Deductions	Net Salary	Password Status	Actions
EMP1001	chandresh Patel	2023040252@vnsgu.ac.in	Information Technology	MERN	\$50000	\$10000	\$5000	\$55000	Bcrypt Encrypted	Edit Delete

Add New Employee (ERP System)

Fill out the employee information below. Password will be encrypted with Bcrypt and emailed via Nodemailer upon saving.

Generated Employee ID:	EMP1002
Auto-Generated Password:	Pass@7107 Will be encrypted using bcrypt before database insertion and emailed to user.
Employee Full Name:	e.g. Rahul Sharma
Employee Email:	e.g. rahul@company.com
Department:	-- Select Department --
Designation:	e.g. Fullstack Developer
Basic Salary (\$):	50000
Allowances (\$):	10000
Deductions (\$):	5000
Calculated Net Salary:	\$55000 (Formula: Basic + Allowances - Deductions)
Save Employee & Send Email Cancel	

Question 6

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localhost:3006

Question 6: Free API Utilities Demonstration

This application demonstrates consuming free public APIs from both the Backend (Node.js/Express) and the Frontend (Browser JavaScript).

1. Utility from BACKEND: Live Currency Converter

How it works: The browser sends a request to the Express server route (/api/currency-convert). The Express backend uses axios to call the free *Open Exchange Rates API*, calculates the converted amount, and returns the result to the user.

Amount to Convert:	100
From Currency (Base):	USD - US Dollar
To Currency (Target):	INR - Indian Rupee
Convert via Backend API	

Conversion Result (Computed by Express Backend):

Original Amount	100 USD
Exchange Rate	1 USD = 95.997282 INR
Converted Amount	9599.73 INR
Data Provider	Open Exchange Rates (via Backend) (Updated: Thu, 01 Oct 2026 00:02:31 +0000)

2. Utility from FRONTEND: Live City Weather Forecast

How it works: The browser executes client-side JavaScript fetch() directly to the free *Open-Meteo Weather API* (no server hop, 100% client-side).

Select City:	New Delhi (India)
Fetch Weather via Frontend API	

Current Weather Details (Direct Browser Fetch):

City Selected	New Delhi (India)
Current Temperature	26.3 °C
Wind Speed	6.3 km/h
Time of Observation	2026-10-01T17:30

Bonus Frontend Utility: Random Joke API

Fetch Random Joke via Frontend API

Setup: What do you call a suspicious looking laptop?
Punchline: Asus

Question 7

←localhost:3007

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User Storefront - Browse Products by Category

Filter by 2-Level Categories:

Level 1 (Parent Category): -- All Categories --

Level 2 (Subcategory): -- All Subcategories --

Available Products

Product Name	Parent Category	Subcategory	Description	Price	Stock	Action
Dell XPS 13 Laptop	Electronics	Laptops & Computers	Intel Core i7, 16GB RAM, 512GB SSD	\$899	10	Add to Cart
Apple iPhone 15	Electronics	Smartphones	128GB Storage, Dynamic Island, A16 Bionic Chip	\$799	25	Add to Cart
Women Floral Summer Dress	Fashion & Clothing	Women Clothing	Elegant casual dress with breathable fabric	\$55	40	Add to Cart
Samsung Galaxy S24	Electronics	Smartphones	8GB RAM, 128GB ROM, AI Camera	\$749	20	Add to Cart
Men Slim Fit Denim Shirt	Fashion & Clothing	Men Clothing	100% Cotton, Casual wear	\$45	50	Add to Cart
Apple MacBook Air M2	Electronics	Laptops & Computers	13-inch liquid retina display, 8GB RAM, 256GB SSD	\$999	15	Add to Cart

Your Shopping Cart

Item	Unit Price	Quantity	Subtotal	Action
Your cart is currently empty.				

Your Shopping Cart

Item	Unit Price	Quantity	Subtotal	Action
Dell XPS 13 Laptop	\$899	- 1 +	\$899	Remove
Total Amount:			\$899	

Proceed to Checkout (\$ 899)

Q2 - Login (File Se... X | Q2 - Protected Ro... X | Q4 - Add New Em... X | Q3 - Protected Ro... X | localhost X | Q5 - Employee E... X | Q6 - Free API Utili... X | Q7 - MERN Shopp... X

←localhost:3007

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MERN Shopping Cart with 2-Level Categories

User Storefront (0 Items)Admin Site (Categories & Products)

Admin Site - Manage 2-Level Categories & Products

1. Add Parent Category (Level 1)

Category Name:

e.g. Sports & Outdoors

Create Parent Category

2. Add Subcategory (Level 2)

Select Parent Category:

-- Select Parent --

Subcategory Name:

e.g. Running Shoes

Create Subcategory

Product Name:

Parent Category (Level 1):

Electronics

Subcategory (Level 2):

Laptops & Computers

Price (\$):

Stock Count:

10

Description:

Save Product

Existing Products in Database

Product Name	Category	Subcategory	Price	Stock	Action
laptop123	Electronics	Laptops & Computers	\$15000	10	Delete
Dell XPS 13 Laptop	Electronics	Laptops & Computers	\$899	10	Delete
Apple iPhone 15	Electronics	Smartphones	\$799	25	Delete
Women Floral Summer Dress	Fashion & Clothing	Women Clothing	\$55	40	Delete
Samsung Galaxy S24	Electronics	Smartphones	\$749	20	Delete
Men Slim Fit Denim Shirt	Fashion & Clothing	Men Clothing	\$45	50	Delete
Apple MacBook Air M2	Electronics	Laptops & Computers	\$999	15	Delete

Existing Categories & Subcategories

Category Name	Level	Parent Category	Action
Electronics	Level 1 (Parent)	None (Top Level)	Delete
Fashion & Clothing	Level 1 (Parent)	None (Top Level)	Delete
Laptops & Computers	Level 2 (Subcategory)	Electronics	Delete
Smartphones	Level 2 (Subcategory)	Electronics	Delete
Men Clothing	Level 2 (Subcategory)	Fashion & Clothing	Delete
Women Clothing	Level 2 (Subcategory)	Fashion & Clothing	Delete
shoes	Level 2 (Subcategory)	Electronics	Delete

Order Placed! Thank you! Order #ORD-1790876709882 placed successfully for \$1898!

Filter by 2-Level Categories:

Level 1 (Parent Category):

-- All Categories --

Level 2 (Subcategory):

-- All Subcategories --

Available Products

Product Name	Parent Category	Subcategory	Description	Price	Stock	Action
laptop123	Electronics	Laptops & Computers	aaaa	\$15000	10	Add to Cart
Dell XPS 13 Laptop	Electronics	Laptops & Computers	Intel Core i7, 16GB RAM, 512GB SSD	\$899	10	Add to Cart
Apple iPhone 15	Electronics	Smartphones	128GB Storage, Dynamic Island, A16 Bionic Chip	\$799	25	Add to Cart
Women Floral Summer Dress	Fashion & Clothing	Women Clothing	Elegant casual dress with breathable fabric	\$55	40	Add to Cart
Samsung Galaxy S24	Electronics	Smartphones	8GB RAM, 128GB ROM, AI Camera	\$749	20	Add to Cart
Men Slim Fit Denim Shirt	Fashion & Clothing	Men Clothing	100% Cotton, Casual wear	\$45	50	Add to Cart
Apple MacBook Air M2	Electronics	Laptops & Computers	13-inch liquid retina display, 8GB RAM, 256GB SSD	\$999	15	Add to Cart

Your Shopping Cart

Item	Unit Price	Quantity	Subtotal	Action
Your cart is currently empty.				

Question 8

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localhost:3008

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Editing student: xyz

Edit Student Details

Roll Number:

101

Student Full Name:

xyz

Email Address:

XYZ@GMAIL.COM

Course / Program:

Computer Science

Total Marks (0-100):

95

Update Student Record

Cancel Edit

Registered Students List (4)

Roll No	Name	Email	Course	Marks	Calculated Grade	Actions
101	xyz	XYZ@GMAIL.COM	Computer Science	95 / 100	A+	EditDelete
DS-103	Rohan Verma	rohan.verma@college.edu	Data Science	68 / 100	B	EditDelete
IT-102	Priya Sharma	priya.sharma@college.edu	Information Technology	92 / 100	A+	EditDelete
CS-101	Aarav Mehta	aarav.mehta@college.edu	Computer Science	88.5 / 100	A+	EditDelete